

Working Title: Architecting Self-Healing MLOps Frameworks for Real-Time Time-Series Forecasting and Concept Drift Adaptation

Objective: To design and deploy resilient, self-healing MLOps architectures that eliminate deployment bottlenecks in real-time predictive pipelines through continuous data integration, automated concept drift monitoring, and dynamic recursive inference.

1. The Context & Problem Statement

While theoretical machine learning models for market prediction and time-series forecasting achieve high baseline accuracy in static experimental settings, they frequently suffer from catastrophic performance degradation when deployed into volatile production environments. In real-world economic systems, sudden macroeconomic shocks - such as fuel price spikes or currency devaluations - trigger rapid **Concept Drift**, rendering static predictive models obsolete. Current production systems lack automated end-to-end telemetry to detect distribution shifts in incoming feature matrices and autonomously trigger model retraining without manual engineering intervention. There is a critical need for robust, virtualization-first MLOps infrastructures capable of continuous monitoring, dynamic schema adaptation, and automated model governance to maintain low-latency inference and prediction accuracy.

2. Proposed Methodology & Technical Architecture

This research investigates the infrastructural and architectural automation of machine learning pipelines using a **Continuous Integration / Continuous Deployment / Continuous Training (CI/CD/CT)** framework. The technical methodology centers on four structural pillars:

- **Automated Data Engineering & Ingestion:** Designing autonomous ingestion layers that extract, parse, and structure high-velocity data feeds while integrating macroeconomic drivers (e.g., exchange rates, fuel indices) and temporal calendar features into a standardized feature store.
- **Non-Parametric Concept Drift Monitoring:** Operationalizing real-time statistical drift monitoring using the two-sample **Kolmogorov-Smirnov (KS) test**. When incoming feature distributions deviate significantly from baseline training statistics ($p < 0.05$) at significance threshold $\alpha = 0.05$, the pipeline automatically raises telemetry alerts and triggers an isolated retraining loop.
- **Dynamic Recursive Forecasting & Model Registry:** Developing unified multi-step recursive inference engines (feeding prediction $\hat{y}_{t+1}$ back as a lagged feature to forecast up to $t+k$) using gradient boosted trees (**XGBoost**), benchmarked continuously against classical autoregressive (**ARIMA**) and linear baselines via **MLflow** model registries.
- **Virtualization-First Serving & Proactive Telemetry:** Containerizing model inference endpoints via **Docker** and **FastAPI** with adaptive input schema detection, coupled with automated CI/CD deployment pipelines and low-latency webhook broadcasting.

3. Preliminary Work & Proof-of-Concept

This research builds directly upon my ongoing MSc dissertation in Data Science at Kingston University, where I have designed and operationalized an end-to-end MLOps pipeline for automated commodity price forecasting within the Sri Lankan retail market.

The foundational proof-of-concept successfully automates daily Central Bank of Sri Lanka (CBSL) report ingestion, drift telemetry, and 7-day recursive XGBoost forecasting within a containerized

Hugging Face Spaces Docker environment.